CYBERSECURITY TRAINING HANDBOOK

SEMESTER 2 (PROFESSIONAL TRACK)

Weeks 27–52 | 6 Days Study (Mon–Sat) | Sunday Revision | Practical First

DOCUMENT CONTROL

Owner:	Yiga Elvis
Document:	Semester 2 Professional Training Handbook
Version:	v1.0 (Generated)
Date:	March 04, 2026
Outcome:	Professional portfolio + job readiness
Policy:	Authorized labs only. No real-world targeting.

1. Purpose (Professional Track)

Semester 2 upgrades you from learner to professional. The objective is job-ready proof: reports, tools, detections/runbooks, and complete case files. Lab-only unless authorized.

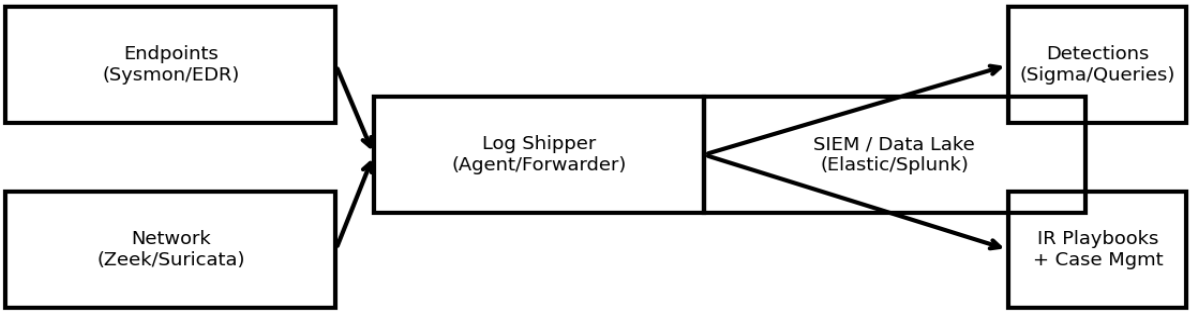
Professional Targets

- 26 weekly artifacts + 26 professional reports minimum.
- Operate a lab with firewall segmentation, AD topology, and SIEM telemetry pipeline.
- Deliver 2 major capstones: Web assessment report + SOC/DFIR case file.
- Produce detection content (Sigma-style) and threat hunt reports with evidence tables.

2. Professional Lab Infrastructure

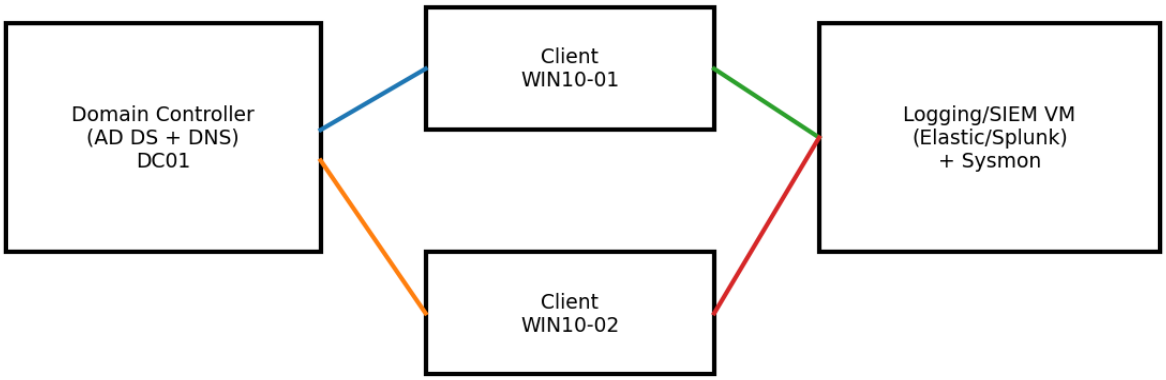
Semester 2 adds firewall segmentation, AD lab, and SIEM pipeline for realistic evidence.

Security Telemetry Pipeline (SOC View)



Visibility → Detections → Response. Weekly evidence + reporting required.

Active Directory Lab (Conceptual Topology)

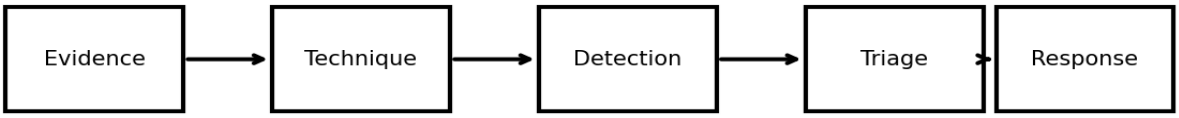


Policy: AD labs used for defense validation and monitoring practice.

3. ATT&CK; Communication (Professional Habit)

Map evidence to technique → detection → triage → response in your reports.

MITRE ATT&CK Mapping Workflow



Professional habit: map findings to ATT&CK and include detections/runbooks.

4. Semester 2 Course Outline (Weeks 27–52)

Month	Week	Theme	Primary Artifact
Month 7	Week 27	Network Security Engineering I: Firewalls & Segmentation (pfSense)	Firewall ruleset + baseline config guide
Month 7	Week 28	Network Security Engineering II: VPN & Secure Remote Access	Remote access policy + DNS/DHCP harden
Month 7	Week 29	Network Telemetry: Zeek/Suricata + PCAP Investigation	PCAP investigation report + indicator table
Month 7	Week 30	Purple Team Validation I (Lab): Scan → Detect → Respond	Validation report + 2 detections + 2 runbooks
Month 8	Week 31	Web Security Engineering I: OWASP Top 10 + Secure Coding	Secure coding checklist + 2 lab writeups
Month 8	Week 32	Web Security Engineering II: Auth, Sessions, JWT, OAuth	Auth testing checklist + 2 lab writeups
Month 8	Week 33	API Security: OWASP API Top 10 + Testing Workflow	API workflow guide + checklist + request col
Month 8	Week 34	Web Capstone: Professional Assessment + Report (Training Targets)	Web assessment report (8–15 pages)
Month 9	Week 35	Active Directory Lab Build: Domain, DNS, GPO (Defensive Focus)	AD build guide + OU/GPO baseline docume
Month 9	Week 36	AD Hardening: Privileged Access + Tiering + Auditing	AD hardening checklist (30 controls)
Month 9	Week 37	Identity Detection (Defense): Kerberos Concepts + Monitoring	Detection idea pack (10) + runbook drafts (3
Month 9	Week 38	Purple Team Validation II (Lab): AD Telemetry + Runbooks	Purple team report + updated runbooks + co
Month 10	Week 39	SIEM Engineering I: Ingest Logs + Query Basics	SIEM notes + 10 useful queries
Month 10	Week 40	Detection Engineering I: Sigma + Tuning	Detection Pack v1 (5 detections + triage step
Month 10	Week 41	Threat Hunting I: Hypothesis-Based Hunts + Reporting	Hunt report pack (2 hunts + gap analysis)
Month 10	Week 42	SOC Simulation: Alert → Investigation → IR Report	SOC case file (2 cases) + IR report (6–10 pa
Month 11	Week 43	Windows DFIR I: Artifact Triage + Timelines	DFIR triage report + timeline table
Month 11	Week 44	Evidence Handling: Hashing + Case File Discipline	Case file template + evidence list with hashe
Month 11	Week 45	Memory Forensics Intro (Training Images): Volatility Workflow	Memory analysis report (3–6 pages)
Month 11	Week 46	Malware Basics (Defensive): Static/Dynamic Concepts + YARA Starter	YARA starter pack (3 rules + results table)
Month 12	Week 47	Cloud Security I: IAM + MFA + Logging Baselines	Cloud baseline doc + logging matrix
Month 12	Week 48	Cloud Security II: Storage + Network Controls + Monitoring	Storage + network security checklist pack
Month 12	Week 49	DevSecOps I: Secrets + Supply Chain + SAST/DAST Concepts	DevSecOps policy pack (checklists + gates)
Month 12	Week 50	Container Security: Docker Baseline Controls (Concept + Practice)	Container baseline checklist + short guide
Month 12	Week 51	Capstone 1: Full Professional Case File (Choose Track)	Capstone case file v1 (10–20 pages + apper
Month 12	Week 52	Capstone 2: Portfolio Packaging + Interview Readiness	Portfolio v1.0 release (README + pinned pr

5. Weekly Modules

Each week includes an overview page and an execution page.

Module: Month 7 — Week 27

Network Security Engineering I: Firewalls & Segmentation (pfSense)

Implement segmentation and firewall policy in a lab; produce a rulebook and hardening report.

Certification alignment: Security+ (controls)	Tools focus: pfSense, Wireshark

Key Notes

- Topics: firewall policy, segmentation, default deny, logging.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: firewall policy, segmentation, default deny, logging.
- Complete labs: pfSense setup + host-only segmentation, Rule creation + logging validation.
- Deliver artifact: Firewall rulebook + baseline config guide.
- Produce scenario report: Write a before/after network hardening report.

Required Artifact

- Firewall rulebook + baseline config guide
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 27

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: firewall policy, segmentation, default deny, logging (risks + defenses).	Start pfSense setup + host-only segmentation.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	pfSense setup + host-only segmentation	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Rule creation + logging validation	Tool log + evidence.
Fri (Practical)	Build artifact.	Firewall rulebook + baseline config guide	Artifact + README.
Sat (Practical)	Scenario + report.	Write a before/after network hardening report	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: pfSense setup + host-only segmentation
- Lab: Rule creation + logging validation
- Build: Firewall rulebook + baseline config guide
- Scenario: Write a before/after network hardening report

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Network Baseline Setup (Authorized)
- Market note: Deliver secure baseline configurations with documented rules and evidence.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- pfSense docs: https://docs.netgate.com/pfsense/en/latest/
- CIS Controls: https://www.cisecurity.org/controls

Module: Month 7 — Week 28

Network Security Engineering II: VPN & Secure Remote Access

Design secure remote access and document policies; understand DNS/DHCP risk controls.

Certification alignment: Security+ (network security)	Tools focus: OpenVPN (optional), dig, Wireshark

Key Notes

- Topics: VPN concepts, remote access policies, DNS/DHCP security.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: VPN concepts, remote access policies, DNS/DHCP security.
- Complete labs: Remote access policy plan, DNS/DHCP hardening checklist + tests.
- Deliver artifact: Remote access policy + DNS/DHCP hardening checklist.
- Produce scenario report: Troubleshoot 'remote user cannot connect' and report actions.

Required Artifact

- Remote access policy + DNS/DHCP hardening checklist
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 28

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: VPN concepts, remote access policies, DNS/DHCP security (risks + defenses).	Start Remote access policy plan.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Remote access policy plan	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	DNS/DHCP hardening checklist + tests	Tool log + evidence.
Fri (Practical)	Build artifact.	Remote access policy + DNS/DHCP hardening checklist	Artifact + README.
Sat (Practical)	Scenario + report.	Troubleshoot 'remote user cannot connect' and report actions	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Remote access policy plan
- Lab: DNS/DHCP hardening checklist + tests
- Build: Remote access policy + DNS/DHCP hardening checklist
- Scenario: Troubleshoot 'remote user cannot connect' and report actions

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Secure Remote Access Guidance (Authorized)
- Market note: Sell a package: secure remote access plan + implementation checklist.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- Cloudflare DNS: https://www.cloudflare.com/learning/dns/
- OpenVPN docs: https://openvpn.net/community-resources/

Module: Month 7 — Week 29

Network Telemetry: Zeek/Suricata + PCAP Investigation

Operate like a network analyst: metadata logs + alerts + PCAP evidence + report writing.

Certification alignment: SOC readiness	Tools focus: Wireshark, Zeek, Suricata

Key Notes

- Topics: Zeek logs, Suricata alerts, PCAP filtering, indicator tables.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: Zeek logs, Suricata alerts, PCAP filtering, indicator tables.
- Complete labs: Zeek analysis on sample traffic, Suricata rule run + alert review.
- Deliver artifact: PCAP investigation report + indicator table.
- Produce scenario report: Write a SOC triage note for suspicious DNS/TLS behavior.

Required Artifact

- PCAP investigation report + indicator table
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 29

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: Zeek logs, Suricata alerts, PCAP filtering, indicator tables (risks + defenses).	Start Zeek analysis on sample traffic.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Zeek analysis on sample traffic	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Suricata rule run + alert review	Tool log + evidence.
Fri (Practical)	Build artifact.	PCAP investigation report + indicator table	Artifact + README.
Sat (Practical)	Scenario + report.	Write a SOC triage note for suspicious DNS/TLS behavior	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Zeek analysis on sample traffic
- Lab: Suricata rule run + alert review
- Build: PCAP investigation report + indicator table
- Scenario: Write a SOC triage note for suspicious DNS/TLS behavior

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: PCAP Investigation Reporting (Authorized)
- Market note: Deliver incident-style reports based on network evidence and recommendations.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- Wireshark: https://www.wireshark.org/docs/
- Zeek: https://zeek.org/
- Suricata: https://suricata.io/

Module: Month 7 — Week 30

Purple Team Validation I (Lab): Scan → Detect → Respond

Validate detections and runbooks using safe lab behaviors; document improvements.

Certification alignment: Detection engineering	Tools focus: Logs, Runbooks

Key Notes

- Topics: telemetry expectations, runbooks, validation, coverage mapping.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: telemetry expectations, runbooks, validation, coverage mapping.
- Complete labs: Generate scan/enum behaviors in lab, Draft 2 detections + 2 runbooks.
- Deliver artifact: Validation report + 2 detections + 2 runbooks (draft).
- Produce scenario report: Tabletop: scan detected on subnet (timeline + actions).

Required Artifact

- Validation report + 2 detections + 2 runbooks (draft)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 30

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: telemetry expectations, runbooks, validation, coverage mapping (risks + defenses).	Start Generate scan/enum behaviors in lab.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Generate scan/enum behaviors in lab	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Draft 2 detections + 2 runbooks	Tool log + evidence.
Fri (Practical)	Build artifact.	Validation report + 2 detections + 2 runbooks (draft)	Artifact + README.
Sat (Practical)	Scenario + report.	Tabletop: scan detected on subnet (timeline + actions)	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Generate scan/enum behaviors in lab
- Lab: Draft 2 detections + 2 runbooks
- Build: Validation report + 2 detections + 2 runbooks (draft)
- Scenario: Tabletop: scan detected on subnet (timeline + actions)

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Detection Validation Exercise (Authorized)
- Market note: Blue teams pay for validation exercises that improve runbooks and detection quality.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- MITRE ATT&CK;: https://attack.mitre.org/
- MITRE D3FEND: https://d3fend.mitre.org/

Module: Month 8 — Week 31

Web Security Engineering I: OWASP Top 10 + Secure Coding

Deepen web security with OWASP Top 10 and practical labs; create secure coding checklist.

Certification alignment: Security+ (app sec)	Tools focus: Burp, DevTools

Key Notes

- Topics: OWASP Top 10, validation/encoding, secure headers.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: OWASP Top 10, validation/encoding, secure headers.
- Complete labs: 2 OWASP training labs, Secure coding checklist for demo app.
- Deliver artifact: Secure coding checklist + 2 lab writeups.
- Produce scenario report: Mini review of a demo site and recommendations.

Required Artifact

- Secure coding checklist + 2 lab writeups
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 31

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: OWASP Top 10, validation/encoding, secure headers (risks + defenses).	Start 2 OWASP training labs.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	2 OWASP training labs	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Secure coding checklist for demo app	Tool log + evidence.
Fri (Practical)	Build artifact.	Secure coding checklist + 2 lab writeups	Artifact + README.
Sat (Practical)	Scenario + report.	Mini review of a demo site and recommendations	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: 2 OWASP training labs
- Lab: Secure coding checklist for demo app
- Build: Secure coding checklist + 2 lab writeups
- Scenario: Mini review of a demo site and recommendations

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Web Security Review (Authorized)
- Market note: Provide checklists + review findings + remediation plan to student/SME projects.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- OWASP Top 10: https://owasp.org/www-project-top-ten/
- OWASP Cheat Sheets: https://cheatsheetseries.owasp.org/
- PortSwigger: https://portswigger.net/web-security

Module: Month 8 — Week 32

Web Security Engineering II: Auth, Sessions, JWT, OAuth

Professional auth: session security, JWT pitfalls, OAuth flow concepts and testing approach.

Certification alignment: Security+ (IAM)	Tools focus: Burp, Postman/curl

Key Notes

- Topics: sessions, JWT, OAuth concepts, auth testing.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: sessions, JWT, OAuth concepts, auth testing.
- Complete labs: JWT training labs, Auth testing checklist (25 items).
- Deliver artifact: Auth testing checklist + 2 lab writeups.
- Produce scenario report: Write a developer advisory: auth mistakes and fixes.

Required Artifact

- Auth testing checklist + 2 lab writeups
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 32

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: sessions, JWT, OAuth concepts, auth testing (risks + defenses).	Start JWT training labs.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	JWT training labs	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Auth testing checklist (25 items)	Tool log + evidence.
Fri (Practical)	Build artifact.	Auth testing checklist + 2 lab writeups	Artifact + README.
Sat (Practical)	Scenario + report.	Write a developer advisory: auth mistakes and fixes	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: JWT training labs
- Lab: Auth testing checklist (25 items)
- Build: Auth testing checklist + 2 lab writeups
- Scenario: Write a developer advisory: auth mistakes and fixes

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Auth Hardening Guidance (Authorized)
- Market note: Sell an auth review package: checklist + evidence + prioritized fixes.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- OWASP Auth Cheat Sheet: https://cheatsheetseries.owasp.org/cheatsheets/Authentication_Cheat_Sheet.html
- OWASP Session Mgmt: https://cheatsheetseries.owasp.org/cheatsheets/Session_Management_Cheat_Sheet.html

Module: Month 8 — Week 33

API Security: OWASP API Top 10 + Testing Workflow

Structured API testing workflow with common risks; produce a professional checklist and mini report.

Certification alignment: Web/API security	Tools focus: Postman, curl

Key Notes

- Topics: OWASP API Top 10, rate limiting, auth headers, logging.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: OWASP API Top 10, rate limiting, auth headers, logging.
- Complete labs: Test a local/demo API, Complete 1 API training lab.
- Deliver artifact: API workflow guide + checklist + request collection.
- Produce scenario report: API mini assessment report (2–4 pages).

Required Artifact

- API workflow guide + checklist + request collection
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 33

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: OWASP API Top 10, rate limiting, auth headers, logging (risks + defenses).	Start Test a local/demo API.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Test a local/demo API	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Complete 1 API training lab	Tool log + evidence.
Fri (Practical)	Build artifact.	API workflow guide + checklist + request collection	Artifact + README.
Sat (Practical)	Scenario + report.	API mini assessment report (2–4 pages)	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Test a local/demo API
- Lab: Complete 1 API training lab
- Build: API workflow guide + checklist + request collection
- Scenario: API mini assessment report (2–4 pages)

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: API Security Checklist Delivery (Authorized)
- Market note: Offer API testing checklists and evidence-based assessments (authorized).
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- OWASP API: https://owasp.org/www-project-api-security/
- MDN HTTP: https://developer.mozilla.org/en-US/docs/Web/HTTP

Module: Month 8 — Week 34

Web Capstone: Professional Assessment + Report (Training Targets)

Full assessment with executive + technical reporting and evidence appendix.

Certification alignment: Reporting	Tools focus: Burp, Screenshots

Key Notes

- Topics: scoping, methodology, findings, remediation.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: scoping, methodology, findings, remediation.
- Complete labs: Assess Juice Shop/DVWA (training), Write findings with evidence.
- Deliver artifact: Web assessment report (8–15 pages).
- Produce scenario report: Executive briefing (1 page) + 5-minute script.

Required Artifact

- Web assessment report (8–15 pages)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 34

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: scoping, methodology, findings, remediation (risks + defenses).	Start Assess Juice Shop/DVWA (training).	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Assess Juice Shop/DVWA (training)	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Write findings with evidence	Tool log + evidence.
Fri (Practical)	Build artifact.	Web assessment report (8–15 pages)	Artifact + README.
Sat (Practical)	Scenario + report.	Executive briefing (1 page) + 5-minute script	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Assess Juice Shop/DVWA (training)
- Lab: Write findings with evidence
- Build: Web assessment report (8–15 pages)
- Scenario: Executive briefing (1 page) + 5-minute script

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Professional Security Reporting (Authorized)
- Market note: Professional reports are a direct money skill (clients pay for clarity + fixes).
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- OWASP WSTG: https://owasp.org/www-project-web-security-testing-guide/
- PortSwigger: https://portswigger.net/web-security

Professional Consolidation (After Week 34) — Checkpoint 1 (After Web Capstone)

Raise quality: redo labs, refactor artifacts, improve reports and evidence.

Mandatory Tasks

- Redo 4 high-value labs (timed; no notes).
- Polish 2 reports to client quality (clear findings + fixes).
- Refactor 2 tools/detections for maintainability.
- Take an 80-question mixed self-test; record weak areas.

Output Required

- Consolidation report (2–4 pages).
- Updated portfolio README + pinned top artifacts.
- Self-test score record + plan.

Module: Month 9 — Week 35

Active Directory Lab Build: Domain, DNS, GPO (Defense Focus)

Build AD lab and document baseline OU/GPO; focus on defense and policy discipline.

Certification alignment: Enterprise readiness	Tools focus: Windows Server (lab), GPO

Key Notes

- Topics: AD DS, DNS, OU/GPO, domain join.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: AD DS, DNS, OU/GPO, domain join.
- Complete labs: Install DC01 + join a client, Create OU and simple GPO baseline.
- Deliver artifact: AD build guide + OU/GPO baseline document.
- Produce scenario report: Write a hardening plan for AD basics.

Required Artifact

- AD build guide + OU/GPO baseline document
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 35

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: AD DS, DNS, OU/GPO, domain join (risks + defenses).	Start Install DC01 + join a client.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Install DC01 + join a client	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Create OU and simple GPO baseline	Tool log + evidence.
Fri (Practical)	Build artifact.	AD build guide + OU/GPO baseline document	Artifact + README.
Sat (Practical)	Scenario + report.	Write a hardening plan for AD basics	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Install DC01 + join a client
- Lab: Create OU and simple GPO baseline
- Build: AD build guide + OU/GPO baseline document
- Scenario: Write a hardening plan for AD basics

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: AD Setup & Documentation (Authorized)
- Market note: Documentation and baseline controls are valuable deliverables in real organizations.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- Microsoft Learn: https://learn.microsoft.com/
- NIST CSF: https://www.nist.gov/cyberframework

Module: Month 9 — Week 36

AD Hardening: Privileged Access + Tiering + Auditing

Design and document hardening controls: privileged access, tiering and audit policies.

Certification alignment: SOC/Enterprise	Tools focus: GPO, Documentation

Key Notes

- Topics: privileged access, admin tiering, audit policies, backups.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: privileged access, admin tiering, audit policies, backups.
- Complete labs: Create AD hardening checklist, Create access review template.
- Deliver artifact: AD hardening checklist (30 controls).
- Produce scenario report: Management brief: top AD controls and why.

Required Artifact

- AD hardening checklist (30 controls)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 36

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: privileged access, admin tiering, audit policies, backups (risks + defenses).	Start Create AD hardening checklist.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Create AD hardening checklist	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Create access review template	Tool log + evidence.
Fri (Practical)	Build artifact.	AD hardening checklist (30 controls)	Artifact + README.
Sat (Practical)	Scenario + report.	Management brief: top AD controls and why	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Create AD hardening checklist
- Lab: Create access review template
- Build: AD hardening checklist (30 controls)
- Scenario: Management brief: top AD controls and why

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: AD Hardening Pack (Authorized)
- Market note: Hardening checklists + implementation support are highly sellable.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- CIS Controls: https://www.cisecurity.org/controls
- Microsoft Learn: https://learn.microsoft.com/

Module: Month 9 — Week 37

Identity Detection (Defense): Kerberos Concepts + Monitoring

Kerberos concepts (defense) and monitoring patterns; produce detection ideas and runbooks.

Certification alignment: Detection engineering	Tools focus: Event logs, Sysmon

Key Notes

- Topics: Kerberos overview, credential hygiene, admin logons, MITRE mapping.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: Kerberos overview, credential hygiene, admin logons, MITRE mapping.
- Complete labs: Collect AD-related logs, Draft 10 detection ideas + 3 runbooks.
- Deliver artifact: Detection idea pack (10) + runbook drafts (3).
- Produce scenario report: Threat hunt report: suspicious admin logons.

Required Artifact

- Detection idea pack (10) + runbook drafts (3)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 37

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: Kerberos overview, credential hygiene, admin logons, MITRE mapping (risks + defenses).	Start Collect AD-related logs.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Collect AD-related logs	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Draft 10 detection ideas + 3 runbooks	Tool log + evidence.
Fri (Practical)	Build artifact.	Detection idea pack (10) + runbook drafts (3)	Artifact + README.
Sat (Practical)	Scenario + report.	Threat hunt report: suspicious admin logons	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Collect AD-related logs
- Lab: Draft 10 detection ideas + 3 runbooks
- Build: Detection idea pack (10) + runbook drafts (3)
- Scenario: Threat hunt report: suspicious admin logons

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Detection Content Pack (Authorized)
- Market note: Sell detection packs and runbooks as a content deliverable to SOC teams.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- MITRE ATT&CK;: https://attack.mitre.org/
- Microsoft Learn: https://learn.microsoft.com/

Module: Month 9 — Week 38

Purple Team Validation II (Lab): AD Telemetry + Runbooks

Validate AD telemetry and improve runbooks; document coverage map and gaps.

Certification alignment: SOC/IR	Tools focus: Logs, Runbooks

Key Notes

- Topics: validation, coverage mapping, false positives, runbooks.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: validation, coverage mapping, false positives, runbooks.
- Complete labs: Run safe AD-related lab tests, Update runbooks + detections.
- Deliver artifact: Purple team report + updated runbooks + coverage map.
- Produce scenario report: IR tabletop: suspicious admin account activity.

Required Artifact

- Purple team report + updated runbooks + coverage map
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 38

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: validation, coverage mapping, false positives, runbooks (risks + defenses).	Start Run safe AD-related lab tests.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Run safe AD-related lab tests	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Update runbooks + detections	Tool log + evidence.
Fri (Practical)	Build artifact.	Purple team report + updated runbooks + coverage map	Artifact + README.
Sat (Practical)	Scenario + report.	IR tabletop: suspicious admin account activity	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Run safe AD-related lab tests
- Lab: Update runbooks + detections
- Build: Purple team report + updated runbooks + coverage map
- Scenario: IR tabletop: suspicious admin account activity

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Purple Team Exercise (Authorized)
- Market note: Professional runbooks reduce response time; organizations pay for them.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- NIST 800-61: https://csrc.nist.gov/publications/detail/sp/800-61/rev-2/final
- MITRE D3FEND: https://d3fend.mitre.org/

Module: Month 10 — Week 39

SIEM Engineering I: Ingest Logs + Query Basics

Ingest telemetry (or samples), write useful queries, and document dashboards.

Certification alignment: SOC readiness	Tools focus: Elastic/Splunk (optional), Sysmon

Key Notes

- Topics: log sources, normalization, queries, dashboards.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: log sources, normalization, queries, dashboards.
- Complete labs: Ingest Sysmon logs, Ingest Zeek or sample dataset.
- Deliver artifact: SIEM notes + 10 useful queries.
- Produce scenario report: Alert triage guide for 3 alert types.

Required Artifact

- SIEM notes + 10 useful queries
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 39

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: log sources, normalization, queries, dashboards (risks + defenses).	Start Ingest Sysmon logs.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Ingest Sysmon logs	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Ingest Zeek or sample dataset	Tool log + evidence.
Fri (Practical)	Build artifact.	SIEM notes + 10 useful queries	Artifact + README.
Sat (Practical)	Scenario + report.	Alert triage guide for 3 alert types	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Ingest Sysmon logs
- Lab: Ingest Zeek or sample dataset
- Build: SIEM notes + 10 useful queries
- Scenario: Alert triage guide for 3 alert types

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: SIEM Query Pack (Authorized)
- Market note: Dashboards and query packs are deliverable outcomes for teams.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- Elastic docs: https://www.elastic.co/guide/
- SigmaHQ: https://github.com/SigmaHQ/sigma

Module: Month 10 — Week 40

Detection Engineering I: Sigma + Tuning

Write quality detections and triage steps; manage false positives.

Certification alignment: Detection engineering	Tools focus: Sigma, Queries

Key Notes

- Topics: Sigma rules, tuning, severity, runbooks.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: Sigma rules, tuning, severity, runbooks.
- Complete labs: Write 5 detections (Sigma-style), Add triage steps and FP notes.
- Deliver artifact: Detection Pack v1 (5 detections + triage steps).
- Produce scenario report: Alert simulation: triage decisions and documentation.

Required Artifact

- Detection Pack v1 (5 detections + triage steps)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 40

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: Sigma rules, tuning, severity, runbooks (risks + defenses).	Start Write 5 detections (Sigma-style).	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Write 5 detections (Sigma-style)	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Add triage steps and FP notes	Tool log + evidence.
Fri (Practical)	Build artifact.	Detection Pack v1 (5 detections + triage steps)	Artifact + README.
Sat (Practical)	Scenario + report.	Alert simulation: triage decisions and documentation	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Write 5 detections (Sigma-style)
- Lab: Add triage steps and FP notes
- Build: Detection Pack v1 (5 detections + triage steps)
- Scenario: Alert simulation: triage decisions and documentation

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Detection Pack Delivery (Authorized)
- Market note: Detection + runbook content is directly monetizable for SOC teams.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- SigmaHQ: https://github.com/SigmaHQ/sigma
- MITRE ATT&CK;: https://attack.mitre.org/

Module: Month 10 — Week 41

Threat Hunting I: Hypothesis-Based Hunts + Reporting

Perform hunts and produce evidence-rich hunt reports (professional format).

Certification alignment: SOC readiness	Tools focus: Queries, Evidence tables

Key Notes

- Topics: baselining, hypothesis hunts, MITRE mapping, reporting.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: baselining, hypothesis hunts, MITRE mapping, reporting.
- Complete labs: Hunt 1: suspicious logons, Hunt 2: suspicious DNS.
- Deliver artifact: Hunt report pack (2 hunts + gap analysis).
- Produce scenario report: Propose 5 new detections based on gaps.

Required Artifact

- Hunt report pack (2 hunts + gap analysis)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 41

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: baselining, hypothesis hunts, MITRE mapping, reporting (risks + defenses).	Start Hunt 1: suspicious logons.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Hunt 1: suspicious logons	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Hunt 2: suspicious DNS	Tool log + evidence.
Fri (Practical)	Build artifact.	Hunt report pack (2 hunts + gap analysis)	Artifact + README.
Sat (Practical)	Scenario + report.	Propose 5 new detections based on gaps	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Hunt 1: suspicious logons
- Lab: Hunt 2: suspicious DNS
- Build: Hunt report pack (2 hunts + gap analysis)
- Scenario: Propose 5 new detections based on gaps

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Threat Hunting Report Service (Authorized)
- Market note: Recurring hunting reports can be delivered monthly (authorized).
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- MITRE ATT&CK;: https://attack.mitre.org/
- NIST CSF: https://www.nist.gov/cyberframework

Module: Month 10 — Week 42

SOC Simulation: Alert → Investigation → IR Report

Run 2 SOC cases and produce full IR reports with timelines and executive brief.

Certification alignment: IR readiness	Tools focus: Timeline building, Reports

Key Notes

- Topics: SOC workflow, IR lifecycle, communication, case documentation.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: SOC workflow, IR lifecycle, communication, case documentation.
- Complete labs: Simulate 2 alert cases, Write IR report with timeline.
- Deliver artifact: SOC case file (2 cases) + IR report (6–10 pages).
- Produce scenario report: Executive briefing (1 page).

Required Artifact

- SOC case file (2 cases) + IR report (6–10 pages)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 42

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: SOC workflow, IR lifecycle, communication, case documentation (risks + defenses).	Start Simulate 2 alert cases.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Simulate 2 alert cases	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Write IR report with timeline	Tool log + evidence.
Fri (Practical)	Build artifact.	SOC case file (2 cases) + IR report (6–10 pages)	Artifact + README.
Sat (Practical)	Scenario + report.	Executive briefing (1 page)	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Simulate 2 alert cases
- Lab: Write IR report with timeline
- Build: SOC case file (2 cases) + IR report (6–10 pages)
- Scenario: Executive briefing (1 page)

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: SOC Case Management (Authorized)
- Market note: Incident handling and reporting quality is a premium professional skill.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- NIST 800-61: https://csrc.nist.gov/publications/detail/sp/800-61/rev-2/final

Professional Consolidation (After Week 42) — Checkpoint 2 (After SOC Simulation)

Raise quality: redo labs, refactor artifacts, improve reports and evidence.

Mandatory Tasks

- Redo 4 high-value labs (timed; no notes).
- Polish 2 reports to client quality (clear findings + fixes).
- Refactor 2 tools/detections for maintainability.
- Take an 80-question mixed self-test; record weak areas.

Output Required

- Consolidation report (2–4 pages).
- Updated portfolio README + pinned top artifacts.
- Self-test score record + plan.

Module: Month 11 — Week 43

Windows DFIR I: Artifact Triage + Timelines

DFIR triage workflow: collect artifacts, build timeline, recommend containment actions.

Certification alignment: DFIR readiness	Tools focus: Sysinternals, PowerShell

Key Notes

- Topics: Windows artifacts, event logs, timeline methods, evidence integrity.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: Windows artifacts, event logs, timeline methods, evidence integrity.
- Complete labs: Collect artifacts in lab, Build timeline table (40 rows).
- Deliver artifact: DFIR triage report + timeline table.
- Produce scenario report: Containment plan based on findings.

Required Artifact

- DFIR triage report + timeline table
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 43

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: Windows artifacts, event logs, timeline methods, evidence integrity (risks + defenses).	Start Collect artifacts in lab.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Collect artifacts in lab	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Build timeline table (40 rows)	Tool log + evidence.
Fri (Practical)	Build artifact.	DFIR triage report + timeline table	Artifact + README.
Sat (Practical)	Scenario + report.	Containment plan based on findings	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Collect artifacts in lab
- Lab: Build timeline table (40 rows)
- Build: DFIR triage report + timeline table
- Scenario: Containment plan based on findings

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: DFIR Triage Service (Authorized)
- Market note: Triage + timeline + recommendations is a strong deliverable package.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- Sysinternals: https://learn.microsoft.com/sysinternals/
- NIST 800-61: https://csrc.nist.gov/publications/detail/sp/800-61/rev-2/final

Module: Month 11 — Week 44

Evidence Handling: Hashing + Case File Discipline

Professional evidence discipline: hashes, documentation, packaging, integrity.

Certification alignment: DFIR readiness	Tools focus: Hashing tools, Documentation

Key Notes

- Topics: evidence handling, hashing, case files, integrity.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: evidence handling, hashing, case files, integrity.
- Complete labs: Create evidence folder with hashes, Integrate evidence list and timeline.
- Deliver artifact: Case file template + evidence list with hashes.
- Produce scenario report: Lessons learned report: evidence quality improvements.

Required Artifact

- Case file template + evidence list with hashes
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 44

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: evidence handling, hashing, case files, integrity (risks + defenses).	Start Create evidence folder with hashes.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Create evidence folder with hashes	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Integrate evidence list and timeline	Tool log + evidence.
Fri (Practical)	Build artifact.	Case file template + evidence list with hashes	Artifact + README.
Sat (Practical)	Scenario + report.	Lessons learned report: evidence quality improvements	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Create evidence folder with hashes
- Lab: Integrate evidence list and timeline
- Build: Case file template + evidence list with hashes
- Scenario: Lessons learned report: evidence quality improvements

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Evidence Collection & Reporting (Authorized)
- Market note: Evidence discipline separates professionals; clients trust clear integrity steps.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- NIST CSF: https://www.nist.gov/cyberframework

Module: Month 11 — Week 45

Memory Forensics Intro (Training Images): Volatility Workflow

Basic memory analysis on training images with structured reporting.

Certification alignment: DFIR readiness	Tools focus: Volatility, Reporting

Key Notes

- Topics: Volatility workflow, processes, connections, confidence levels.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: Volatility workflow, processes, connections, confidence levels.
- Complete labs: Analyze training memory image, Extract process list + suspicious indicators.
- Deliver artifact: Memory analysis report (3–6 pages).
- Produce scenario report: ATT&CK; mapping of findings (concept).

Required Artifact

- Memory analysis report (3–6 pages)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 45

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: Volatility workflow, processes, connections, confidence levels (risks + defenses).	Start Analyze training memory image.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Analyze training memory image	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Extract process list + suspicious indicators	Tool log + evidence.
Fri (Practical)	Build artifact.	Memory analysis report (3–6 pages)	Artifact + README.
Sat (Practical)	Scenario + report.	ATT&CK; mapping of findings (concept)	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Analyze training memory image
- Lab: Extract process list + suspicious indicators
- Build: Memory analysis report (3–6 pages)
- Scenario: ATT&CK; mapping of findings (concept)

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Memory Analysis Reporting (Authorized)
- Market note: Memory analysis reporting is a premium DFIR skill (authorized).
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- Volatility: https://www.volatilityfoundation.org/
- MITRE ATT&CK;: https://attack.mitre.org/

Module: Month 11 — Week 46

Malware Basics (Defensive): Static/Dynamic Concepts + YARA Intro

Safe malware analysis concepts and starter YARA rules for detection.

Certification alignment: Detection/DFIR	Tools focus: YARA, Hashing

Key Notes

- Topics: malware types, static indicators, dynamic concepts, YARA basics.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: malware types, static indicators, dynamic concepts, YARA basics.
- Complete labs: Write 3 simple YARA rules (training), Test rules on safe samples.
- Deliver artifact: YARA starter pack (3 rules + results table).
- Produce scenario report: Malware analysis summary template (1–2 pages).

Required Artifact

- YARA starter pack (3 rules + results table)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 46

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: malware types, static indicators, dynamic concepts, YARA basics (risks + defenses).	Start Write 3 simple YARA rules (training).	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Write 3 simple YARA rules (training)	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Test rules on safe samples	Tool log + evidence.
Fri (Practical)	Build artifact.	YARA starter pack (3 rules + results table)	Artifact + README.
Sat (Practical)	Scenario + report.	Malware analysis summary template (1–2 pages)	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Write 3 simple YARA rules (training)
- Lab: Test rules on safe samples
- Build: YARA starter pack (3 rules + results table)
- Scenario: Malware analysis summary template (1–2 pages)

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Defensive Detection Pack (Authorized)
- Market note: Detection content (YARA) + reporting is sellable to SOC teams.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- YARA: https://virustotal.github.io/yara/
- CISA: https://www.cisa.gov/

Module: Month 12 — Week 47

Cloud Security I: IAM + MFA + Logging Baselines

Cloud baseline thinking: shared responsibility, IAM least privilege, MFA, and logging.

Certification alignment: Cloud readiness	Tools focus: IAM concepts, JSON/YAML

Key Notes

- Topics: shared responsibility, IAM, MFA, logging matrix.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: shared responsibility, IAM, MFA, logging matrix.
- Complete labs: Create baseline checklist, Draft logging matrix (source/retention/alerts).
- Deliver artifact: Cloud baseline doc + logging matrix.
- Produce scenario report: Mini playbook: leaked API keys response.

Required Artifact

- Cloud baseline doc + logging matrix
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 47

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: shared responsibility, IAM, MFA, logging matrix (risks + defenses).	Start Create baseline checklist.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Create baseline checklist	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Draft logging matrix (source/retention/alerts)	Tool log + evidence.
Fri (Practical)	Build artifact.	Cloud baseline doc + logging matrix	Artifact + README.
Sat (Practical)	Scenario + report.	Mini playbook: leaked API keys response	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Create baseline checklist
- Lab: Draft logging matrix (source/retention/alerts)
- Build: Cloud baseline doc + logging matrix
- Scenario: Mini playbook: leaked API keys response

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Cloud Baseline Review (Authorized)
- Market note: Cloud baseline reviews are high-demand professional services.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- AWS Security: https://docs.aws.amazon.com/security/
- Azure Security: https://learn.microsoft.com/azure/security/

Module: Month 12 — Week 48

Cloud Security II: Storage + Network Controls + Monitoring

Protect data in cloud: storage security and network controls; produce baseline checklists.

Certification alignment: Cloud readiness	Tools focus: Architecture diagrams, Checklists

Key Notes

- Topics: storage security, encryption concepts, network controls, monitoring.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: storage security, encryption concepts, network controls, monitoring.
- Complete labs: Write storage checklist, Design segmentation concept.
- Deliver artifact: Storage + network security checklist pack.
- Produce scenario report: Cloud architecture briefing (1 page).

Required Artifact

- Storage + network security checklist pack
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 48

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: storage security, encryption concepts, network controls, monitoring (risks + defenses).	Start Write storage checklist.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Write storage checklist	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Design segmentation concept	Tool log + evidence.
Fri (Practical)	Build artifact.	Storage + network security checklist pack	Artifact + README.
Sat (Practical)	Scenario + report.	Cloud architecture briefing (1 page)	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Write storage checklist
- Lab: Design segmentation concept
- Build: Storage + network security checklist pack
- Scenario: Cloud architecture briefing (1 page)

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Cloud Architecture Review (Authorized)
- Market note: Architecture reviews with checklists are sellable to many teams.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- CIS Controls: https://www.cisecurity.org/controls
- NIST CSF: https://www.nist.gov/cyberframework

Module: Month 12 — Week 49

DevSecOps I: Secrets + Supply Chain + SAST/DAST Concepts

Secure delivery pipeline thinking: secrets hygiene, dependency risk, and security gates.

Certification alignment: Security engineering	Tools focus: Git, Policy writing

Key Notes

- Topics: secrets, dependency risks, SAST/DAST, SBOM concept.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: secrets, dependency risks, SAST/DAST, SBOM concept.
- Complete labs: Secrets hygiene checklist, Pipeline gate policy draft.
- Deliver artifact: DevSecOps policy pack (checklists + gates).
- Produce scenario report: Apply checklist to sample repo and write findings.

Required Artifact

- DevSecOps policy pack (checklists + gates)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 49

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: secrets, dependency risks, SAST/DAST, SBOM concept (risks + defenses).	Start Secrets hygiene checklist.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Secrets hygiene checklist	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Pipeline gate policy draft	Tool log + evidence.
Fri (Practical)	Build artifact.	DevSecOps policy pack (checklists + gates)	Artifact + README.
Sat (Practical)	Scenario + report.	Apply checklist to sample repo and write findings	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Secrets hygiene checklist
- Lab: Pipeline gate policy draft
- Build: DevSecOps policy pack (checklists + gates)
- Scenario: Apply checklist to sample repo and write findings

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: DevSecOps Checklist Service (Authorized)
- Market note: Teams pay for secure release checklists and pipeline policies.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- OWASP ASVS: https://owasp.org/www-project-application-security-verification-standard/
- OWASP Cheat Sheets: https://cheatsheetseries.owasp.org/

Module: Month 12 — Week 50

Container Security: Docker Baseline Controls (Concept + Practice)

Baseline container security: images, runtime least privilege, logging and hardening checklist.

Certification alignment: Cloud/DevSecOps	Tools focus: Docker, Checklists

Key Notes

- Topics: Docker basics, image risks, runtime least privilege, logging.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: Docker basics, image risks, runtime least privilege, logging.
- Complete labs: Run a safe container locally, Draft container baseline checklist.
- Deliver artifact: Container baseline checklist + short guide.
- Produce scenario report: Incident scenario note: container compromise response.

Required Artifact

- Container baseline checklist + short guide
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 50

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: Docker basics, image risks, runtime least privilege, logging (risks + defenses).	Start Run a safe container locally.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Run a safe container locally	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Draft container baseline checklist	Tool log + evidence.
Fri (Practical)	Build artifact.	Container baseline checklist + short guide	Artifact + README.
Sat (Practical)	Scenario + report.	Incident scenario note: container compromise response	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Run a safe container locally
- Lab: Draft container baseline checklist
- Build: Container baseline checklist + short guide
- Scenario: Incident scenario note: container compromise response

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Container Baseline Guidance (Authorized)
- Market note: Container baseline documents are useful deliverables for dev teams.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- Docker docs: https://docs.docker.com/
- CIS Docker Benchmark: https://www.cisecurity.org/benchmark/docker

Module: Month 12 — Week 51

Capstone 1: Full Professional Case File (Choose Track)

Deliver a full professional case file with evidence, report, and recommendations.

Certification alignment: Professional readiness	Tools focus: Reporting, Evidence management

Key Notes

- Topics: scoping, risk rating, report writing, presentation.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: scoping, risk rating, report writing, presentation.
- Complete labs: Choose track: Web/SOC/AD/Cloud (lab), Produce full report + appendix.
- Deliver artifact: Capstone case file v1 (10–20 pages + appendix).
- Produce scenario report: 1-page executive brief + 5-minute briefing script.

Required Artifact

- Capstone case file v1 (10–20 pages + appendix)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 51

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: scoping, risk rating, report writing, presentation (risks + defenses).	Start Choose track: Web/SOC/AD/Cloud (lab).	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Choose track: Web/SOC/AD/Cloud (lab)	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Produce full report + appendix	Tool log + evidence.
Fri (Practical)	Build artifact.	Capstone case file v1 (10–20 pages + appendix)	Artifact + README.
Sat (Practical)	Scenario + report.	1-page executive brief + 5-minute briefing script	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Choose track: Web/SOC/AD/Cloud (lab)
- Lab: Produce full report + appendix
- Build: Capstone case file v1 (10–20 pages + appendix)
- Scenario: 1-page executive brief + 5-minute briefing script

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Professional Consulting Pack (Authorized)
- Market note: Capstone case files are the strongest portfolio assets for hiring.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- NIST 800-61: https://csrc.nist.gov/publications/detail/sp/800-61/rev-2/final
- MITRE ATT&CK;: https://attack.mitre.org/

Module: Month 12 — Week 52

Capstone 2: Portfolio Packaging + Interview Readiness

Finalize portfolio, CV bullet bank, interview pack, and 90-day plan.

Certification alignment: Job readiness	Tools focus: GitHub, Professional communication

Key Notes

- Topics: portfolio packaging, CV bullets, STAR stories, ethics/law basics.
- Write professional evidence: objective, method, evidence, findings, remediation.
- Mon–Tue theory only; Wed–Sat labs and reporting.

Learning Outcomes

- Demonstrate: portfolio packaging, CV bullets, STAR stories, ethics/law basics.
- Complete labs: Pin top artifacts, Build interview pack (40 Q&As;).
- Deliver artifact: Portfolio v1.0 release (README + pinned projects + CV bullets).
- Produce scenario report: Mock interview: present one case file and one detection pack.

Required Artifact

- Portfolio v1.0 release (README + pinned projects + CV bullets)
- Evidence + README required.
- Report (2–6 pages) required.

Execution Plan — Week 52

2 theory days maximum; practical + reporting dominates.

Daily Plan (Mon–Sat study | Sun revision)

Day	Theory Focus	Hands-On Lab	Deliverable / Proof
Mon (Theory)	Theory: portfolio packaging, CV bullets, STAR stories, ethics/law basics (risks + defenses).	Start Pin top artifacts.	Notes + evidence.
Tue (Theory)	Theory: detection/testing + remediation approach.	Write checklist + 10-question quiz.	Checklist + score.
Wed (Practical)	Lab 1 (core).	Pin top artifacts	Writeup + evidence.
Thu (Practical)	Lab 2 (tool mastery).	Build interview pack (40 Q&As;)	Tool log + evidence.
Fri (Practical)	Build artifact.	Portfolio v1.0 release (README + pinned projects + CV bullets)	Artifact + README.
Sat (Practical)	Scenario + report.	Mock interview: present one case file and one detection pack	Report (2–6 pages).
Sun (Revision)	Revision + self-test.	Redo 1 lab + 30-question test.	Score + weak areas.

Lab Tasks (Required)

- Lab: Pin top artifacts
- Lab: Build interview pack (40 Q&As;)
- Build: Portfolio v1.0 release (README + pinned projects + CV bullets)
- Scenario: Mock interview: present one case file and one detection pack

Exercises

- Write 1-page executive brief and 1-page technical brief.
- Create a junior-analyst checklist for the week.
- Add 15 Q&As; to your revision notebook.

Platform Targets

- TryHackMe/HTB: 1 aligned module.
- If web: 1 PortSwigger lab.

Money-Generating Concept (Authorized)

- Skill: Portfolio-to-Job Conversion (Authorized)
- Market note: Portfolio packaging converts skill into paid roles and contracts.
- Start: Offer authorized: assessment + evidence + report + remediation checklist.

References

- GitHub Docs: https://docs.github.com/
- NIST CSF: https://www.nist.gov/cyberframework

Appendix A — Detection Rule / Runbook Template

Field	Guidance
Title	Short detection name.
Data Sources	Which logs are required.
Logic	Rule/query logic in clear terms.
False Positives	Expected benign patterns + tuning.
Severity	Critical/High/Medium/Low with rationale.
Triage Steps	What analyst checks first.
Response	Containment steps and escalation path.
Mapping	MITRE technique mapping (if applicable).

Appendix B — Threat Hunt Report Template

Section	What to write
Hypothesis	Testable statement.
Data Sources	Logs + time range.
Queries	Exact queries/filters and why.
Findings	Evidence table with timestamps.
Conclusion	Confirmed/not confirmed; confidence.
Recommendations	New detections, tuning, hardening actions.

Appendix C — Incident Report Template

Section	What to write
Executive Summary	What happened + impact + status.
Timeline	Timestamps + evidence + actions.
Detection	How detected.
Scope	Systems/accounts/data.
Containment	Actions to stop impact.
Eradication	Removal steps.
Recovery	Restore/monitor/validate.
IOCs	Domains/IPs/hashes.
Lessons Learned	Improvements and controls.
Appendix	Evidence exports/hashes.

Appendix D — Vulnerability Finding Template

Section	What to write
Title	Short, specific finding name.
Severity	Critical/High/Medium/Low + reason.
Affected Asset	Host/IP/app endpoint (authorized scope).
Evidence	Screenshots/commands/PCAP/log references.
Impact	What could happen; map to CIA.
Root Cause	Misconfiguration / code flaw / missing control.
Remediation	Fix steps + prevention guidance.
Verification	Confirm fix (before/after).